A HR Application based on Force.com PaaS

CLOUD COMPUTING AND SOFTWARE AS A SERVICE ASSIGNMENT 2

Subject Coordinator:
Professor Farookh Khadeer Hussain

Members:
Duong Kham Dang - 24999322
Meiling Dong - 14537827
Jiwei Jiang - 25633037

Content List

Introduction 2

Data Model 2

Custom user profile 3

Workflow diagrams 6

Prototype System Screens and Description 10

Conclusion and Future Work 14

Contribution sheet 16

Reference list 17

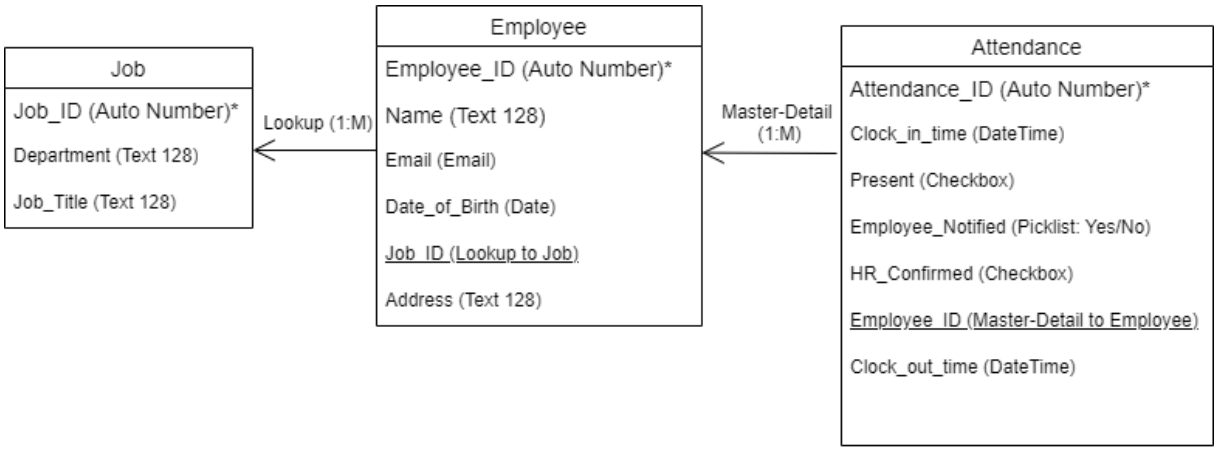
Introduction

This project was commissioned by the human resources department of a medium-sized enterprise to develop an efficient and scalable Human Resources Management System (HRMS) using the Force.com platform to meet its key business needs in intelligent rationalization of employee attendance management. With the continuous expansion of enterprise scale, traditional paper or local systems are no longer able to support flexible permission configuration, cross departmental collaboration, and real-time data access. Therefore, the CEO and CIO of the enterprise decided to try introducing cloud-based PaaS solutions and quickly build a verifiable prototype system using Salesforce's Force.com platform. This platform not only supports data object modeling, business logic configuration, and automated processes, but also provides powerful security mechanisms and flexible user access control, making it suitable for deployment in enterprise level HR scenarios (Saleforce 2022). In this project, we will focus on developing four core functions: "automatically record employees' clock in and clock out information", "timely notify employees and HR through email and task reminders", "set validation rules such as age limit and mandatory position", "HR can uniformly manage and confirm the punch in information of all employees" and "realize user authority isolation and sensitive field protection", and demonstrate how the Force.com platform can meet the current and future human resource management needs of enterprises through a validated prototype.

System Modelling

Data Model

Below is our data model.



A job can correspond to multiple employees, one employee corresponds to multiple attendance records, employee associates jobs through the lookup field. An employee can have multiple attendance records. Deleting an employee will delete the corresponding attendance records because of Master-Detail relationship.

Custom user profile

We create three profiles: CEO (Admin), Employee and Human Resource Manager.

Analytics Cloud Integration User	Analytics Cloud Integration User	☐
Analytics Cloud Security User	Analytics Cloud Integration User	☐
Anypoint Integration	Identity	☐
Authenticated Website	Authenticated Website	☐
Authenticated Website	Authenticated Website	☐
CEO Profile	Salesforce	✓
Chatter External User	Chatter External	☐
Chatter Free User	Chatter Free	☐
Chatter Moderator User	Chatter Free	☐
Contract Manager	Salesforce	☐
Cross Org Data Proxy User	XOrg Proxy User	☐
Customer Community Login User	Customer Community Login	☐
Customer Community Plus Login User	Customer Community Plus Login	☐
Customer Community Plus User	Customer Community Plus	☐
Customer Community User	Customer Community	☐
Customer Portal Manager Custom	Customer Portal Manager Custom	☐
Customer Portal Manager Standard	Customer Portal Manager Standard	☐
Einstein Agent User	Einstein Agent	☐
Employee Profile	Salesforce	✓
External Apps Login User	External Apps Login	☐
External Identity User	External Identity	☐
Force.com - App Subscription User	Force.com - App Subscription	☐
Profile Name ↑	**User License**	**Custom**
High Volume Customer Portal User	High Volume Customer Portal	☐
Human Resource Manager	Salesforce	✓
Identity User	Identity	☐
Marketing User	Salesforce	☐
Minimum Access - API Only Integrations	Salesforce Integration	☐
Minimum Access - Salesforce	Salesforce	☐
Partner App Subscription User	Partner App Subscription	☐
Partner Community Login User	Partner Community Login	☐
Partner Community User	Partner Community	☐
Read Only	Salesforce	✓
Salesforce API Only System Integrations	Salesforce Integration	✓
Silver Partner User	Silver Partner	☐
Solution Manager	Salesforce	☐
Standard Platform User	Salesforce Platform	☐
Standard User	Salesforce	☐
System Administrator	Salesforce	☐
Work.com Only User	Work.com Only	☐

Employee - Employee can only view and modify their employee records (such as address and mailbox). Employee can only view and create own clock in records, and employee cannot modify or access other people's information

CEO (system admin) - The CEO has full authority over all objects, fields and records, and manages all settings such as workflow and validation rules, field permissions, etc.

Human Resource Manager - HR can view and edit employee and attendance records of all employees - can receive tasks and confirm clock in records (HR_confirmed field).

Our methods to realize data access rights and other functions include: using custom profiles to restrict object access (object & tab level security), and using sharing settings to set the employee object to private to avoid mutual visits between employees Set master detail for attendance to make the record belong to the current user. Use field level security to hide sensitive fields such as birth date Use validation rules to restrict illegal data, such as under 18 years old or unfilled position.

We have three objectives: Attendance, Employee and Job. We create different validation rules for the objectives to keep data safety and achieve some functions.

1. Attendance

We set below rules: Clock-in time must be within 5 minutes of the current time, Clock-out time must be within 5 minutes of the current time, employee cannot repeat clock-in and clock-out, clock-in time cannot be later than clock-out time.

Validation Rules — Validation Rules Help

Rule Name	Error Location	Error Message	Active	Modified By
Attendance_ClockIn_5Min_Tolerance	Clock-in time	Clock-in time must be within 5 minutes of the current time. Please refresh and try again.	✓	Meiling Dong, 5/7/2025, 9:27 AM
Attendance_ClockOut_5Min_Tolerance	Clock-out time	Clock-out time must be within 5 minutes of the current time. Please refresh and try again.	✓	Meiling Dong, 5/7/2025, 9:31 AM
Prevent_Duplicate_Clock_In	Clock-in time	Do not repeat clock in!	✓	Meiling Dong, 5/7/2025, 10:04 AM
Prevent_Duplicate_Clock_Out	Clock-out time	Do not repeat clock out!	✓	Meiling Dong, 5/7/2025, 10:03 AM
Require_Clock_In_Before_Clock_Out	Clock-out time	Clock-in time cannot be later than Clock-out time.	✓	Meiling Dong, 5/7/2025, 10:19 AM

2. Employee

We set below rules: Date of birth cannot be in the future, employees must be at least 18 years old.

Validation Rules — Validation Rules Help

Rule Name	Error Location	Error Message	Active	Modified By
Birthdate_Cannot_Be_Future	Address	Date of birth cannot be in the future.	✓	Meiling Dong, 5/5/2025, 12:52 PM
Minimum_Age_18	Address	Employees must be at least 18 years old.	✓	Meiling Dong, 5/5/2025, 12:45 PM

3. Job

We set below rules: Job Title cannot be blank.

Validation Rules — Validation Rules Help

Rule Name	Error Location	Error Message	Active	Modified By
Job_Title_Required	Department	Job Title cannot be blank.	✓	Meiling Dong, 5/5/2025, 12:48 PM

Object-level security

Custom Object Permissions

	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records	Modify All Records	View All Fields
Attendances	✓	✓	✓	☐	☐	☐	☐
Employees	✓	☐	✓	☐	☐	☐	☐

	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records	Modify All Records	View All Fields
Jobs	✓	☐	☐	☐	☐	☐	☐

Job Field-Level Security for profile
Employee Profile — Help for this Page

Edit | Back to Profile

Field Name	Field Type	Read Access	Edit Access
Created By	Lookup	✓	☐
Department	Text	✓	✓
Job ID	Auto Number	✓	☐
Job Title	Text	✓	✓
Last Modified By	Lookup	✓	☐
Owner	Lookup	✓	✓

Edit | Back to Profile

Attendance Field-Level Security for profile
Employee Profile — Help for this Page

Edit | Back to Profile

Field Name	Field Type	Read Access	Edit Access
Attendance ID	Auto Number	✓	☐
Clock-in time	Date/Time	✓	✓
Clock-out time	Date/Time	✓	✓
Created By	Lookup	✓	☐
Employee	Lookup	✓	✓
Employee Notified	Picklist	✓	☐
HR Confirmed	Checkbox	✓	☐
Last Modified By	Lookup	✓	☐
Present	Checkbox	✓	✓

Edit | Back to Profile

Employee Field-Level Security for profile
Employee Profile — Help for this Page

Edit | Back to Profile

Field Name	Field Type	Read Access	Edit Access
Address	Text	✓	✓
Created By	Lookup	✓	☐
Date of Birth	Date	✓	✓
Email	Email	✓	✓
Employee_Job_Lookup_Relationship	Lookup	✓	☐
Employee ID	Auto Number	✓	☐
Last Modified By	Lookup	✓	☐
Name	Text	✓	☐
Owner	Lookup	✓	✓

Edit | Back to Profile

Sharing setting

Attendance	Controlled by Parent	Controlled by Parent	
Employee	Private	Private	✓
Job	Public Read Only	Public Read Only	✓

Organisation-wide default

Organization-Wide Defaults — Organization-Wide Defaults Help

Object	Default Internal Access	Default External Access	Grant Access Using Hierarchies
Attendance	Controlled by Parent	Controlled by Parent	

Tab-level security

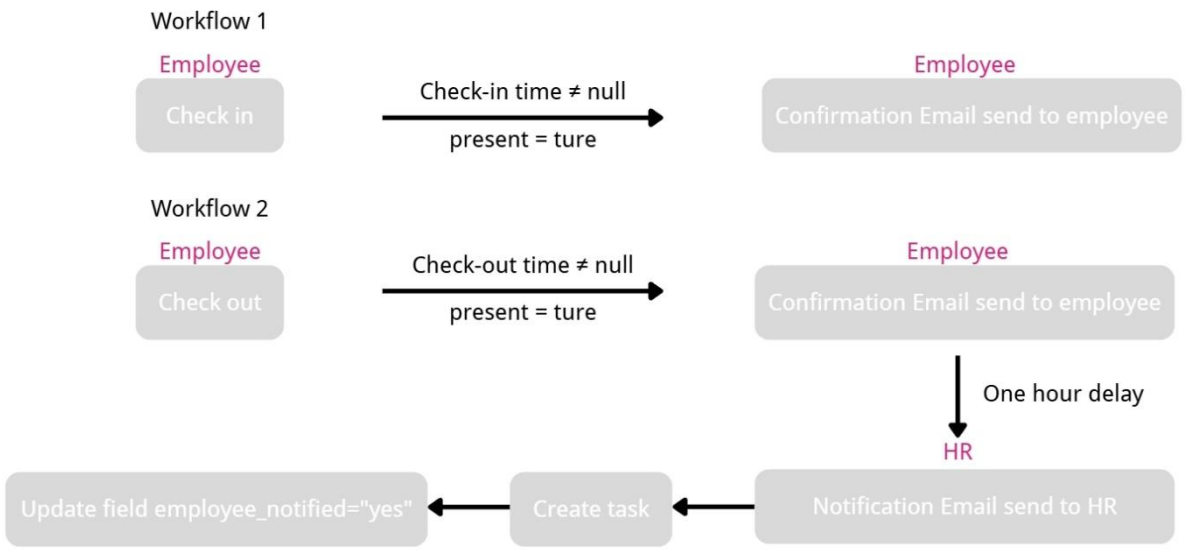
Custom Tab Settings

Attendances	Default On	Jobs	Default On
Employees	Default On	Welcome	Default On

Custom App Settings

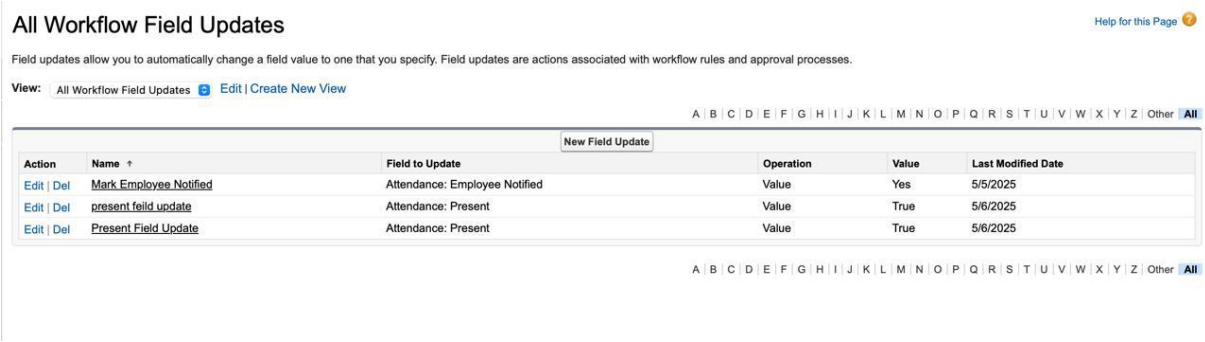
	Visible	Default
All Tabs (standard__AllTabSet)	☐	☐
Analytics Studio (standard__Insights)	☐	☐
Approvals (standard__Approvals)	☐	☐
Automation (standard__FlowsApp)	☐	☐
Bolt Solutions (standard__LightningBolt)	☐	☐
Community (standard__Community)	☐	☐
Content (standard__Content)	☐	☐
Data Manager (standard__DataManager)	☐	☐
Developer Edition (Get_Started_with_Agentforce)	☐	☐
Digital Experiences (standard__SalesforceCMS)	☐	☐
HR Attendance System (HR_Attendance_System)	☑	☑
Human Resource Management (Human_Resource_Management)	☐	☐
Lightning Usage App (standard__LightningInstrumentation)	☐	☐

Workflow diagrams



Workflow 1: enter the specific check in time when the employee clicks check in. After that, employees will receive a successful check in email.

Workflow2: enter the specific check-out time when the employee checks out. After that, employees will receive a successful check-out time. At the same time, create a task to HR. HR will receive a reminder email after one hour and confirm the attendance of employees.



All Workflow Field Updates

Help for this Page

Field updates allow you to automatically change a field value to one that you specify. Field updates are actions associated with workflow rules and approval processes.

View: All Workflow Field Updates Edit | Create New View

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All

New Field Update

Action	Name	Field to Update	Operation	Value	Last Modified Date
Edit | Del	Mark Employee Notified	Attendance: Employee Notified	Value	Yes	5/5/2025
Edit | Del	present feild update	Attendance: Present	Value	True	5/6/2025
Edit | Del	Present Field Update	Attendance: Present	Value	True	5/6/2025

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All

All Workflow Rules

Help for this Page

Go with the flow! With Flow Builder, the future of low-code automation, you can do everything you do with workflow rules — and more! Salesforce plans to retire workflow rules and recommends building automation in Flow Builder. Tell Me More | Migrate your workflow rules to flows

Configure your organization's workflow by creating workflow rules. Each workflow rule consists of:

- Criteria that cause the workflow rule to run.
- Immediate actions that execute when a record matches the criteria. For example, Salesforce can automatically send an email that notifies the account team when a new high-value opportunity is created.
- Time-dependent actions that queue when a record matches the criteria, and execute according to time triggers. For example, Salesforce can automatically send an email reminder to the account team if a high-value opportunity is still open ten days before the close date.

Quick Tips
- Useful Sample Workflow Rule
- Video Tutorial (English Only)
- Troubleshooting Workflow

View: All Workflow Rules

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All

Rule Name	Description	Object	Active
Clock_in_Email	Rule for checking if an employee is present	Attendance	✓
Send_Clock_Out_Notifications	Check clock out time is not null	Attendance	✓

Workflow Rule
Clock_in_Email

Help for this Page

« Back to List: Workflow Rules

Go with the flow! With Flow Builder, the future of low-code automation, you can do everything you do with workflow rules — and more! Salesforce plans to retire workflow rules and recommends building automation in Flow Builder. Tell Me More | Migrate your workflow rules to flows

Workflow Rule Detail

Rule Name	Clock_in_Email	Object	Attendance
Active	✓	Evaluation Criteria	Evaluate the rule when a record is created
Description	Rule for checking if an employee is present		
Rule Criteria	`Attendance: Clock-in time` NOT EQUAL TO null		
Created By	Melling Dong, 5/5/2025, 4:47 PM	Modified By	Duong Kham Dang, 5/6/2025, 12:44 AM

Workflow Actions

Immediate Workflow Actions

Type	Description
Email Alert	Send Clock-in Confirmation
Field Update	Present Field Update

Workflow Rule
Send_Clock_Out_Notifications

Help for this Page

« Back to List: Workflow Rules

Go with the flow! With Flow Builder, the future of low-code automation, you can do everything you do with workflow rules — and more! Salesforce plans to retire workflow rules and recommends building automation in Flow Builder. Tell Me More | Migrate your workflow rules to flows

Workflow Rule Detail

Rule Name	Send_Clock_Out_Notifications	Object	Attendance
Active	✓	Evaluation Criteria	Evaluate the rule when a record is created, and any time it's edited to subsequently meet criteria
Description	Check clock out time is not null		
Rule Criteria	`Attendance: Clock-out time` NOT EQUAL TO null		
Created By	Melling Dong, 5/5/2025, 5:54 PM	Modified By	Duong Kham Dang, 5/6/2025, 12:40 AM

Workflow Actions

Immediate Workflow Actions

Type	Description
Email Alert	Send Clock-out Confirmation
Field Update	Mark Employee Notified

Time-Dependent Workflow Actions See an example

1 Hour After Attendance: Clock-out time

Type	Description
Task	Confirm Clock-out Record
Email Alert	Notify HR to verify clock-out

Below are our three email notifications and task setting.

All Email Alerts

Help for this Page

Email alerts are used to send emails from a flow or other automation.

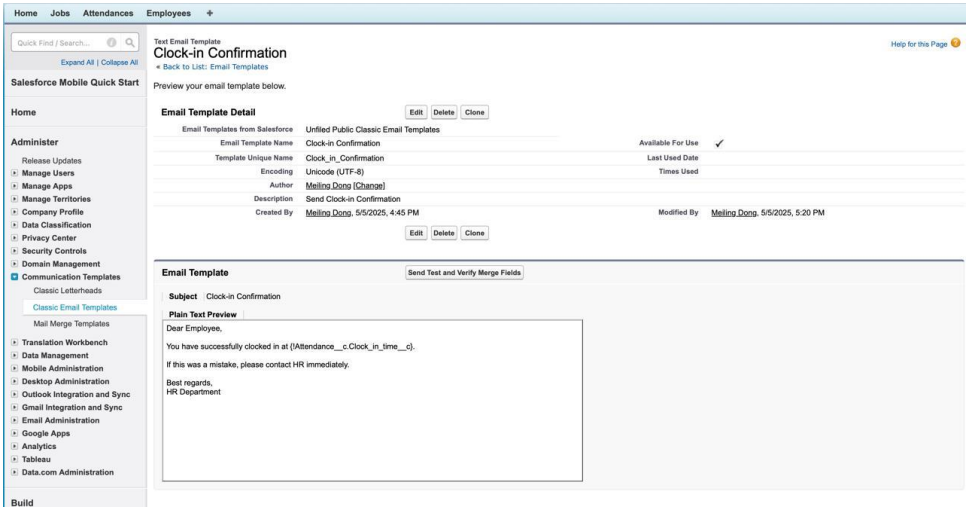
View: All Email Alerts Create New View

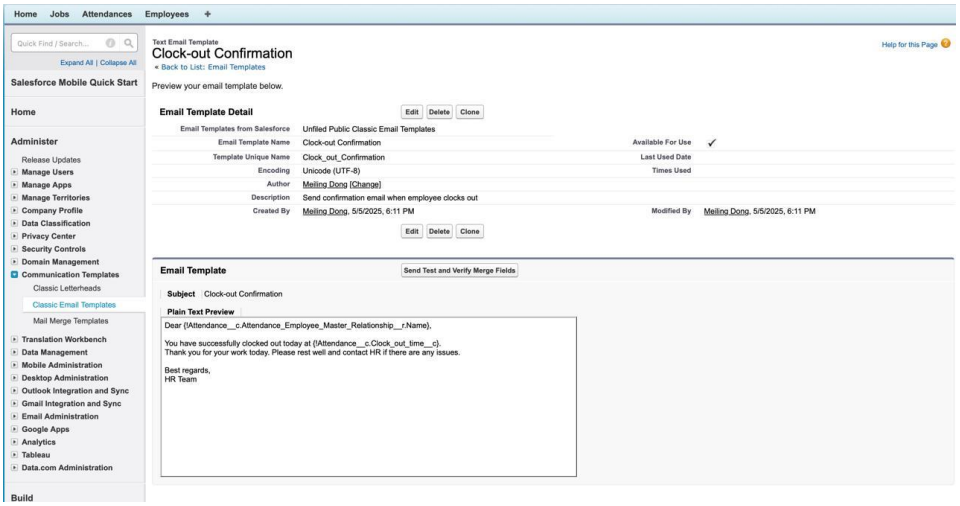
A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All

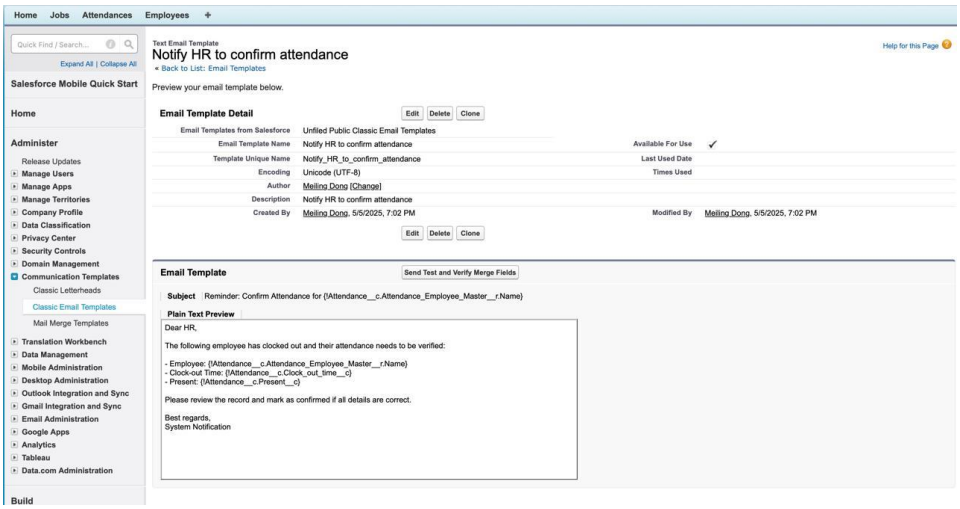
New Email Alert

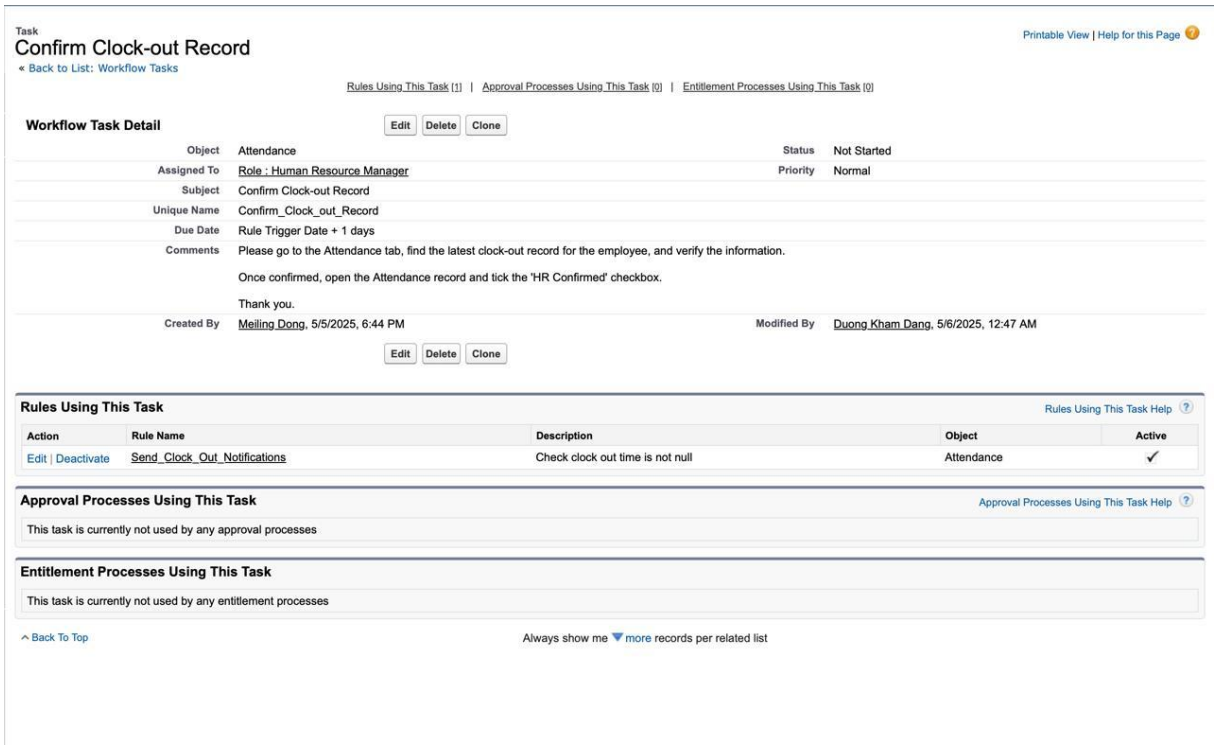
Action	Description	Email Template Name	Object	Last Modified Date
Edit | Del	Notify HR to verify clock-out	Notify HR to confirm attendance	Attendance	5/5/2025
Edit | Del	Send Clock-in Confirmation	Clock-in Confirmation	Attendance	5/7/2025
Edit | Del	Send Clock-out Confirmation	Clock-out Confirmation	Attendance	5/7/2025

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All


Home Jobs Attendances Employees
Text Email Template
Clock-in Confirmation
Back to List: Email Templates
Help for this Page
Preview your email template below.
Email Template Detail
Edit Delete Clone
Email Templates from Salesforce Unfiled Public Classic Email Templates
Email Template Name Clock-in Confirmation
Available For Use
Template Unique Name Clock_in_Confirmation
Last Used Date
Encoding Unicode (UTF-8)
Times Used
Author Meiling Dong [Change]
Description Send Clock-in Confirmation
Created By Meiling Dong, 5/5/2025, 4:45 PM
Modified By Meiling Dong, 5/5/2025, 5:20 PM
Email Template
Send Test and Verify Merge Fields
Subject Clock-in Confirmation
Plain Text Preview
Dear Employee,
You have successfully clocked in at {!Attendance__c.Clock_in_time__c}.
If this was a mistake, please contact HR immediately.
Best regards,
HR Department

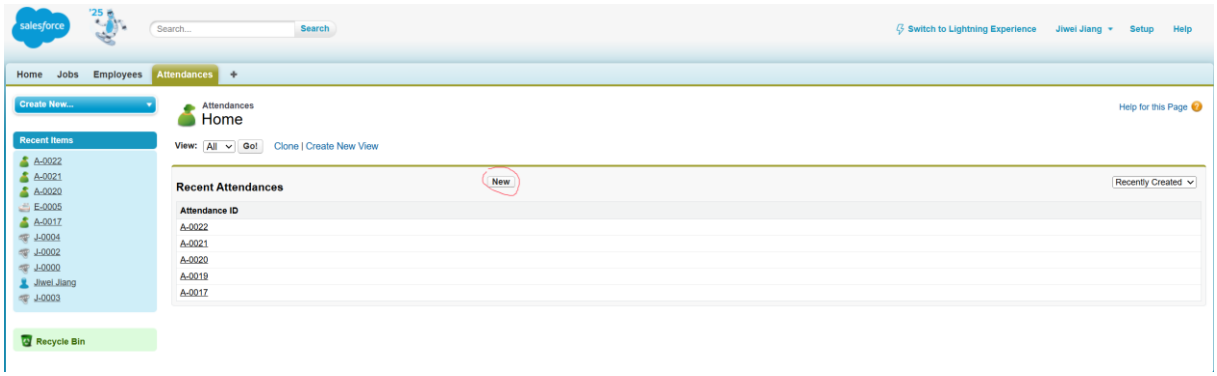

Home Jobs Attendances Employees
Text Email Template
Clock-out Confirmation
Back to List: Email Templates
Help for this Page
Preview your email template below.
Email Template Detail
Edit Delete Clone
Email Templates from Salesforce Unfiled Public Classic Email Templates
Email Template Name Clock-out Confirmation
Available For Use
Template Unique Name Clock_out_Confirmation
Last Used Date
Encoding Unicode (UTF-8)
Times Used
Author Meiling Dong [Change]
Description Send confirmation email when employee clocks out
Created By Meiling Dong, 5/5/2025, 6:11 PM
Modified By Meiling Dong, 5/5/2025, 6:11 PM
Email Template
Send Test and Verify Merge Fields
Subject Clock-out Confirmation
Plain Text Preview
Dear {!Attendance__c.Attendance_Employee_Master_Relationship__r.Name},
You have successfully clocked out today at {!Attendance__c.Clock_out_time__c}.
Thank you for your work today. Please rest well and contact HR if there are any issues.
Best regards,
HR Team


Home Jobs Attendances Employees
Text Email Template
Notify HR to confirm attendance
Back to List: Email Templates
Help for this Page
Preview your email template below.
Email Template Detail
Edit Delete Clone
Email Templates from Salesforce Unfiled Public Classic Email Templates
Email Template Name Notify HR to confirm attendance
Available For Use
Template Unique Name Notify_HR_to_confirm_attendance
Last Used Date
Encoding Unicode (UTF-8)
Times Used
Author Meiling Dong [Change]
Description Notify HR to confirm attendance
Created By Meiling Dong, 5/5/2025, 7:02 PM
Modified By Meiling Dong, 5/5/2025, 7:02 PM
Email Template
Send Test and Verify Merge Fields
Subject Reminder: Confirm Attendance for {!Attendance__c.Attendance_Employee_Master__r.Name}
Plain Text Preview
Dear HR,
The following employee has clocked out and their attendance needs to be verified:
- Employee: {!Attendance__c.Attendance_Employee_Master__r.Name}
- Clock-out Time: {!Attendance__c.Clock_out_time__c}
- Present: {!Attendance__c.Present__c}
Please review the record and mark as confirmed if all details are correct.
Best regards,
System Notification

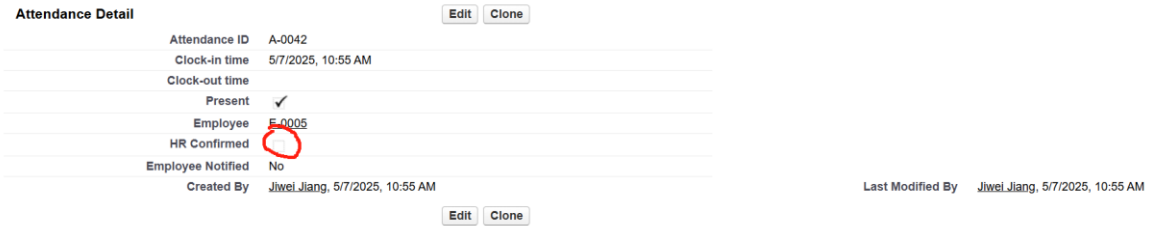


Prototype System Screens and Description

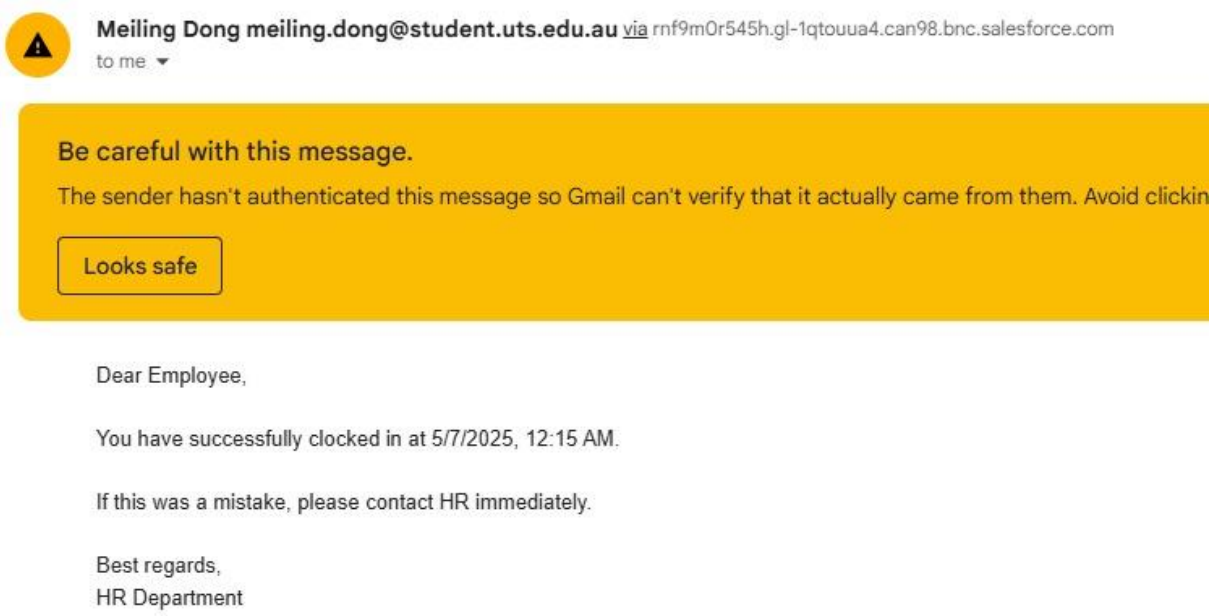
I am an employee of this company and I want to check-in. The first step is click new to create a new attendance.



Then input check-in details and save. Because of validation rules, employees can only select the clock in time within five minutes before and after the current time. They can only check-in and then check-out. And employees cannot repeat check-in.



For now, the HR confirmed is unticked. Once check-in successfully, the confirmation will be sent to the employee's email.



As same as check-in, we do the same thing to check-out

Attendance Detail Edit Clone

Attendance ID	A-0042
Clock-in time	5/7/2025, 10:55 AM
Clock-out time	5/7/2025, 10:57 AM
Present	✓
Employee	E-0005
HR Confirmed	☐
Employee Notified	Yes
Created By	Jiwei Jiang, 5/7/2025, 10:55 AM

Once check-out successfully, the confirmation will be sent to the employee's email.

Meiling Dong meiling.dong@student.uts.edu.au via ssexdzbnac57.gl-1qtouua4.can98.bnc.salesforce.com
to me

Be careful with this message.

The sender hasn't authenticated this message so Gmail can't verify that it actually came from them. Avo

Looks safe

Dear ,

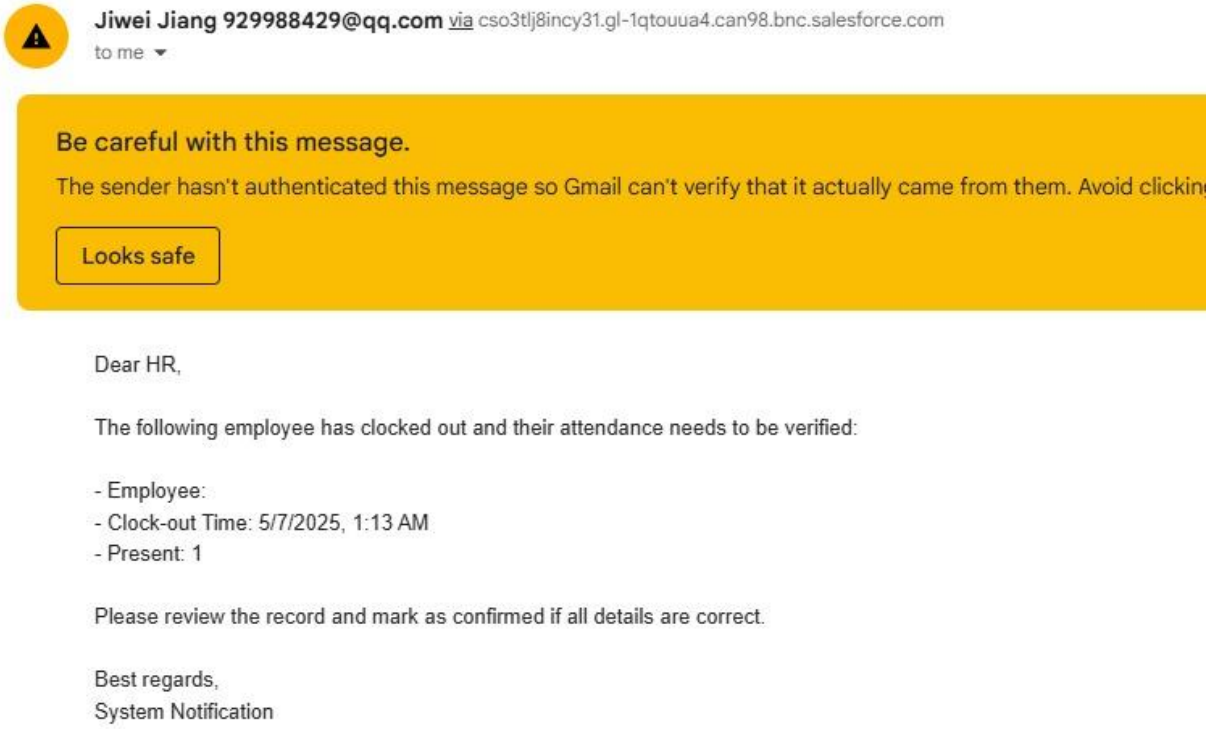
You have successfully clocked out today at 5/7/2025, 12:21 AM.
Thank you for your work today. Please rest well and contact HR if there are any issues.

Best regards,
HR Team

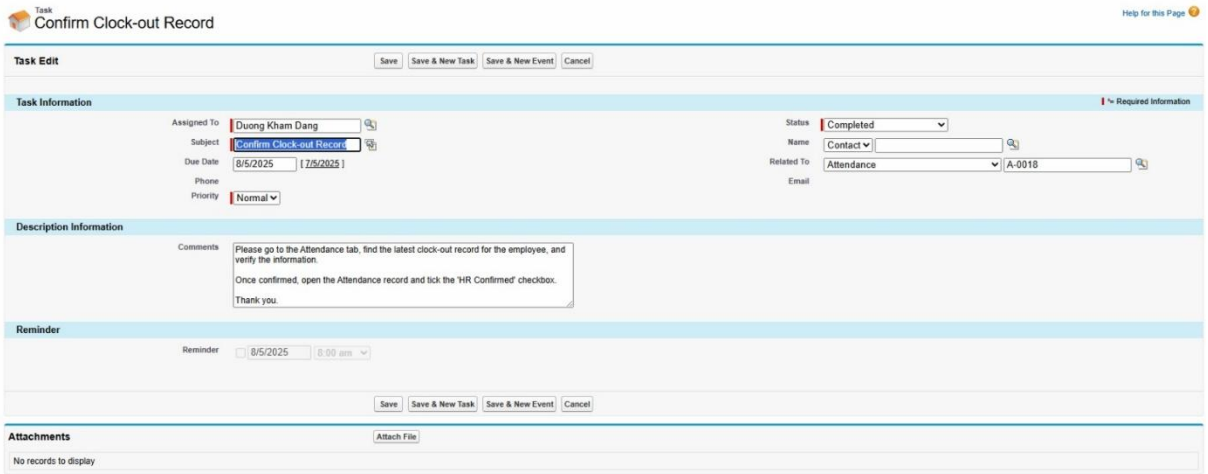
After one hour from the employee check-out, the HR will get a task and one notification email will be sent to HR's mail box.

My Tasks New All Open

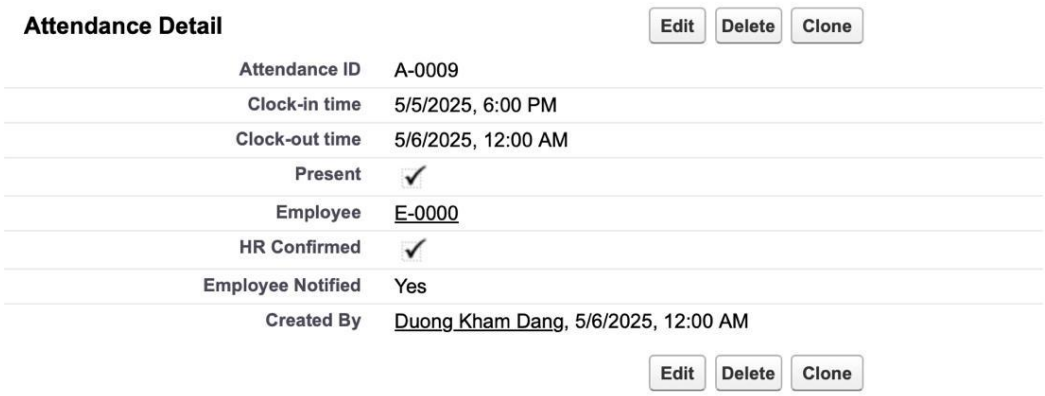
Complete	Date	Status	Subject	Name	Related To	Account
x	8/5/2025	Not Started	Confirm Clock-out Record		A-0016	
x	8/5/2025	Not Started	Confirm Clock-out Record		A-0017	
x	8/5/2025	Not Started	Confirm Clock-out Record		A-0018	
x	8/5/2025	Not Started	Confirm Clock-out Record		A-0020	
x	8/5/2025	Not Started	Confirm Clock-out Record		A-0019	



Then, HR can change the status of the check-in and check-out. All the attendance will be finished.



Then the attendance HR confirmed will be ticked.



Attendance Detail

Edit Delete Clone

Attendance ID	A-0009
Clock-in time	5/5/2025, 6:00 PM
Clock-out time	5/6/2025, 12:00 AM
Present	✓
Employee	E-0000
HR Confirmed	✓
Employee Notified	Yes
Created By	Duong Kham Dang, 5/6/2025, 12:00 AM

Edit Delete Clone

Conclusion and Future Work

Our project has basically realized the automation of employee attendance and the security and privacy of employee files. Through the development of HR management system based on force.com platform, we have deeply experienced the software development method based on PAAS (platform as a service). Compared with traditional development, force.com provides a highly configurable data object modeling tool, a flexible authority management mechanism, and powerful workflow and automation functions, enabling us to build a prototype system with core HR functions in a short time. In addition, the visual interface and logical configuration of the platform reduce the development threshold, so that team members can effectively cooperate and complete the system design even with limited programming experience (Zhang et al. 2020).

In the development process, we have mastered the key function modules such as object relationship configuration (such as lookup and master detail), data validation rule settings, user-defined role creation, organization level permission configuration, automatic email notification, task allocation and time triggered workflow. These functions not only improve the stability and automation of the system, but also lay a good foundation for the expansion and integration of the system in the future.

Looking forward to the future, the HR system prototype can be further expanded into a complete fully automatically human resource management platform, covering modules such as salary management, performance appraisal, training records, employee self-service portal, etc. In addition, the force.com platform can also be applied to other scenarios within the enterprise, such as contract management, customer support, supply chain management, etc. (Saleforce 2025), so as to realize the digitization and automation of business processes in a wider range. Based on the experience of this prototype development, we believe that force.com is an ideal choice for enterprises to carry out rapid and agile development and realize cloud business process optimization.

Contribution sheet

Contributor	Contributions
Duong Kham Dang	Develop the Data Model
Duong Kham Dang	Implement Data Validation Rules
Duong Kham Dang	Configure Object-level Security
Duong Kham Dang	Configure Tab-Level Security
Meiling Dong	Implement Email Notifications
Meiling Dong	Configure Task Creation and Assignment
Meiling Dong	Implement Time-Based Workflow
Jiwei Jiang	Create Workflow and Data Model Diagrams
Jiwei Jiang	Report writing
All the team member	App Test
All the team member	Brainstorm
All the team member	System Modelling Screens
All the team member	Final Review

Force.com Login Credential:

Username: dangduongkham1703156@agentforce.com

Password: Assignment2

Reference list

Salesforce 2022, ‘Salesforce’, *Salesforce*, viewed 5 May 2025, <https://www.salesforce.com/uk/service/employee-engagement/>.

Salesforce 2025, ‘Developer Documentation | Salesforce Developers’, *developer.salesforce.com*, viewed <https://developer.salesforce.com/docs>.

Zhang, Q., Cheng, L. & Boutaba, R. 2020, ‘Cloud computing: state-of-the-art and research challenges’, *Journal of Internet Services and Applications*, vol. 1, no. 1, pp. 7–18.